Week 3 — Factors, Multiples & Number Patterns — Answer Key

Q1. C) 6 — 18 = 6 × 3, so 6 is a factor. (4, 5, 8 do not divide 18 exactly.)

Q2. B) 35 — 35 = 7 × 5. (30, 40, 45 are not multiples of 7.)

Q3. B) 6 — Factors of 12: 1, 2, 3, 4, 6, 12. Factors of 18: 1, 2, 3, 6, 9, 18. Greatest common factor = 6.

Q4. B) 12 — Multiples of 4: 4, 8, 12, 16... Multiples of 6: 6, 12, 18... First common one = 12.

Q5. C) 29 — 29 has no factors other than 1 and itself. (21 = 3×7, 27 = 3×3×3, 33 = 3×11.)

Q6. C) 6 — Factors of 12: 1, 2, 3, 4, 6, 12 — that's 6 factors.

Q7. C) 15 — 15 = 3 × 5 is the smallest number that both 3 and 5 divide exactly.

Q8. C) 48 — 9 × 5 = 45, 9 × 6 = 54, so 48 falls between multiples of 9 and isn't one itself. (27 = 9×3, 36 = 9×4, 63 = 9×7.)

Q9. C) 48 — Each number is double the one before it (×2): 3, 6, 12, 24, 48.

Q10. B) 70 — Each number decreases by 10: 100, 90, 80, 70, 60.

Q11. C) 36 — These are square numbers: 1^2, 2^2, 3^2, 4^2, 5^2, 6^2 = 1, 4, 9, 16, 25, 36.

Q12. B) 9 — These are odd numbers increasing by 2: 1, 3, 5, 7, 9, 11.

Q13. C) 32 — Each number is double the one before it: 2, 4, 8, 16, 32, 64.

Q14. B) 7 groups, 0 left — 42 ÷ 6 = 7 exactly, so 7 full groups with nothing left over.

Q15. B) 8 — 8 is a factor of 24 (24 ÷ 8 = 3) and also a multiple of 4 (4 × 2 = 8). (6 is a factor of 24 but not a multiple of 4; 10 and 16 are not even factors of 24.)

Scoring: 15/15 — excellent, move to Week 4. 13–14/15 — solid, review the missed ones. Below 13/15 — go back over factors vs. multiples (they're the most commonly confused pair at this stage) before Week 4.